COMPANY

Technology Services Provider
solution proposal for
Global Automotive Manufacturer
Parts Inspection Data Management System

Technology Services Provider / Global Automotive Manufacturer
PI DMS



COMPANY LOGO

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A. Document Control

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1. Version Control

Date	Document Version Name/ Number	Author	Verified By
8th Oct 2026	Technology Services Provider Solution Proposal for Global Automotive Manufacturer - Data Management System v1.0	Technology Services Provider Solution Team	Technology Services Provider Business Team

B. Executive Summary

1. Cover Letter

Hi Global Automotive Manufacturer Team,

Thank you for giving us this opportunity to participate in meeting your business requirements for your Data Management System for Global Automotive Manufacturer. The proposed solution has been carefully formulated, incorporating the insights gained from our discussions and the information shared by the Global Automotive Manufacturer Team. It is crucial to engage in further discussions with the key stakeholders of your Team to ensure the effective adoption of use-case-specific strategies and address any potential need for customization services.

Technology Services Provider looks forward to working with you and your team on this business engagement and to a long and mutually rewarding business relationship. Should you have any questions regarding this proposal response, do feel free to connect with the respective team members mentioned in Section L of this proposal document.

Yours sincerely,
Team Digital Engineering, (Technology Services Provider)

2. Our Brief Understanding of the Business Requirement

Global Automotive Manufacturer aims to digitize its current paper-driven Quality & Inspection processes - covering Audits, SMIR/SIS-P Validation, Regulatory Parts Inspection and PACD into an integrated, cloud-based digital platform. It will replace email-driven and paper-based processes with structured workflows, real-time validations, approvals, and a secure document repository.

Technology Services Provider proposes to design and develop a secure, scalable, and user-friendly web platform with tablet-based inspection capability. The solution will cover:

- Regulatory Parts Inspection (Tablet Compatible Android App) with automated reporting
- Audit planning, execution, reporting, and approvals
- Monthly SMIR Validation & Digital Approval Workflows
- Digitize the PACD process end-to-end
- Dashboards, real-time alerts, centralized repository, and audit trails

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C. Scope of Work

1. Project Objectives

The primary objectives of this project are:

- Digitize audit-related processes across the PI Department.
- Enable Tablet-Based Inspection of Regulatory Parts.
- To design and implement a unified digital system that streamlines Parts Quality processes by incorporating SISP approval workflows and monthly SMIR validation within a single platform.
- Streamline corrective action management by replacing manual PACD workflows with a governed, system-driven process.
- Create a Unified, Cloud-Based Repository.
- Improve traceability, compliance, and transparency.
- Reduce turnaround time, errors, and paper usage.

2. Functional Modules

The processes will be implemented according to the steps detailed below:

A. Regulatory Parts Inspection (via Android App)

Provide inspectors a tablet-friendly android mobile app to perform regulatory part checks on the shop floor. The app must enforce inspection sequence, capture structured observations and evidence (photos) and produce automated inspection reports that flow into the central DMS for approvals and analytics.

- The PPC team shares daily/monthly production plan sync and assigns inspection tasks to inspectors in the new system.
- Inspector receives task list in system and visits Assembly line as per production plan.
- The inspector will then perform inspection as per the checksheets provided in the system by the Homologation department.
- The inspection will be carried out on random samples taken from the trolley or bin in sequence. All observations must be recorded in the digital checksheet using the appropriate input type which can be text notes, validated numeric fields (min/max/tolerance), dropdown selections, image capture, or file uploads.
- If the result from the inspection is NG, then the concerned department(Homologation, VI and PI) will be notified through email and alerts with NG details. Only sharing the details of NG parts is enough, no follow up needed.
- If the inspection results are OK, the report is automatically forwarded to the division head or the designated approval hierarchy for acknowledgement. Once acknowledged, the checksheet is marked as complete and stored digitally in DMS for future reference.

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Currently, around 40 checksheets are created each month and they are to be managed by two users using tablets.

B. Audit Planning & Scheduling

Digitize end-to-end audit operations of the PI department which includes planning, execution, evidence capture, improvement tracking, multi-level approvals, and centralized document management. Removing manual paperwork and enabling faster, transparent, and compliant decision-making.

- Enable PI users to create and manage Annual and Monthly Audit Plans within the DMS for defined categories:
  - Worst Supplier Audit
  - QAV Audit
  - Special Process Supplier Visit
  - System & Process Audit
- Allow users to capture key audit details including Supplier, Audit Category, Frequency, Month, and Tentative Audit Date.
- Automatically map and attach standardized checksheets based on the selected audit category as defined by Global Automotive Manufacturer.
- Provide Admin/SuperAdmin control to modify checksheet fields (add/remove fields only; no changes to graphs or visual elements).
- Display planned audits in the assigned auditor's task list for execution.
- Enable auditors to perform audits digitally, record observations directly in the system, and attach supporting evidence (photos, documents, files).
- Allow auditors to submit completed checksheets through the DMS, after which the system will lock the record to prevent changes (with controlled Admin/SuperAdmin override).
- Restrict internal visibility of checksheets to Global Automotive Manufacturer users, while generating a separate Improvement Report for supplier sharing.
- Enable suppliers to submit action plans against improvement feedback within five working days through the system.
- Support iteration workflows, where unsatisfactory action plans are returned to suppliers for revision via automated alerts.
- Allow auditors to create a cover note once the action plan is approved.
- Automatically compile the audit package, including the final checksheet, approved action plan, and supporting evidence.
- Implement a multi-level approval workflow (4-5 levels), including Department Head, Operational Head and Division Head.
- Allow approvers to approve or request clarifications before finalization.
- Upon approval, mark the cover note as final and store all documents in a centralized

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repository.

- Ensure all records remain searchable and available for future audits, traceability, and compliance reviews.

C. SIS-P and SMIR Management

A new module will be developed within the DMS to enable supplier-driven SIS-P creation and monthly SMIR submission.

- Digitize the complete SIS-P lifecycle by enabling creation, submission, review, and approval of SIS-P documents directly within the application. The SIS-P approval process will involve multiple stakeholders, with the approval matrix to be defined and provided by Global Automotive Manufacturer.
- Enable end-to-end SMIR processing by allowing suppliers to submit SMIRs through the system, with automated validation and review by internal teams.
- Eliminate PDF and email dependency by storing SIS-P data in structured digital formats within the database.
- Automate validation and comparison between SIS-P and SMIR using system rules, improving accuracy, consistency, and audit readiness.
- Migrate legacy SIS-P documents (~4000 records) using OCR powered by AI Document Intelligence to extract data from existing PDFs. The extracted data will be stored in a structured database format to enable automated comparison and validation in the new system.
- Create a single source of truth for all SIS-P and SMIR records, enabling faster reviews, historical traceability, and future analytics.
- Supplier performs inspection at their end in line with SIS-P as per production plan and uploads monthly summary inspection report to the DMS portal before the 7th of every month.
- Implement a rule-based SMIR validation engine to automatically check:
  - Completeness of supplier data
  - Tolerance deviations (OK/NG detection)
  - Sample size vs production plan
  - Logical validations (NG, abnormal spikes)
- The responsible PI member will review the SMIR, and the system will automatically highlight any findings based on predefined validations, enabling quicker and more accurate decision-making.
- If the SMIR is marked as NG, it will be rejected and an automated email with remarks will be sent to the supplier. The new system will track re-uploaded corrected SMIRs through version control (v1, v2, v3 etc).

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- If the SMIR is marked as OK, it will be digitally approved along with the approver's details and timestamp, and then stored in the application.

Note: Currently, Global Automotive Manufacturer works with approximately 70 suppliers who submit SIS-P and SMIR documents, as of now ~4000 SIS-P records have to be digitized. The text extraction from SIS-P PDFs is a one time activity to be performed by Technology Services Provider for ~4000 SIS-P records only.

D. PACD (Problem Analysis and Corrective Document)

- Digitize the PACD process by enabling PI teams to create PACD forms directly in the system when defective parts are identified, eliminating email-based communication.
- Enable direct supplier response by allowing suppliers to fill, update,and submit PACD forms within the application.
- Automate follow-ups and reminders to ensure timely supplier submissions and reduce manual tracking.
- Support root cause documentation by allowing the 4M diagram to be uploaded as an image (diagram creation remains external to the system).
- Enforce supplier approvals by routing completed PACD forms to the supplier's authorized approver before final submission.
- Allow evidence attachment through image uploads to support corrective actions.
- Track turnaround timelines by providing a 15-day submission window from PACD initiation to supplier response.
- Enable structured internal approvals with a three-level Global Automotive Manufacturer approval workflow before PACD closure.
- Ensure traceability and audit readiness by maintaining a complete digital record of PACD history, evidence, and approvals.

E. Reporting & Dashboarding

- The reports will be generated based on the data captured through the new DMS.
- Reports will be filterable and exportable in multiple formats (Excel, CSV, DOC, and PDF).
- Access rights for viewing and generating reports will be configurable based on user roles.
- The dashboard interface will be user-friendly, plant-specific, and provide visual summaries for quick insights.

F. User Manuals Documentation

- Technology Services Provider will prepare comprehensive text and video based User Manuals for both End Users and Super Admin / Admin, detailing step-by-step instructions on how to operate the new DMS.

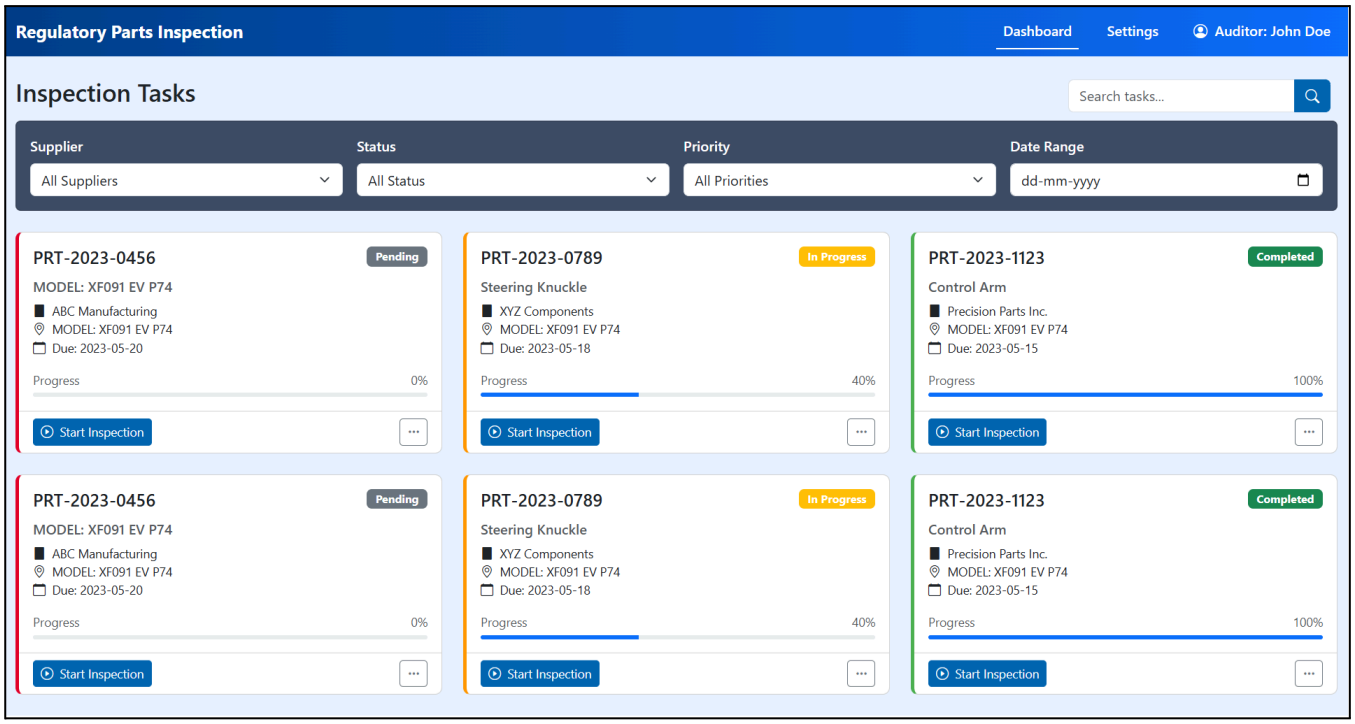
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G. User and Role Management

- Administrators can add new users, edit user details, or deactivate/remove users as required.
- The module will include search and filter capabilities to easily manage large user bases.
- Administrators can assign/revoke roles to individual users.

D. Prototypes

1. Regulatory Parts Inspection



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2. Parts Identification Checksheet

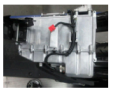
Regulatory Parts Inspection — Dashboard | Settings | Auditor: John Doe

Parts Identification Checksheet

Export | Submit Inspection

Part Information	Supplier	Inspector	Status
PRT-2023-0456 MODEL: XF091 EV P74	ABC Manufacturing MODEL: XF091 EV P74	John Doe 2023-05-15	In Progress Sample: 5 units

Dimensional Checks

S.NO.	Components	Manufacturer Name	Trade Name / Trade Mark on Component	Part ID	Part ID (Observed)	Full Part	Master	Audit	Remarks
1	Battery	Suzuki Motor Corp.	Suzuki Motor Corp.	73100-32L00	73100-32L00				OK
2	Battery Management System	RNTEC	RNTEC	L3B1605A	L3B1605A				OK
3	DC-DC Converter	Uno Minda Limited	Uno Minda Limited	34DC360002-0010E0 / 72780-32L00	34DC360002-0010E0 / 72780-32L00				OK
4	Charger	Spark Minda Green Mobility Systems Pvt. Ltd.	Spark Minda Green Mobility Systems Pvt. Ltd.	73750-32L	73750-32L				OK
5	VCU	BOSCH	BOSCH	72510-32L00	72510-32L00				OK

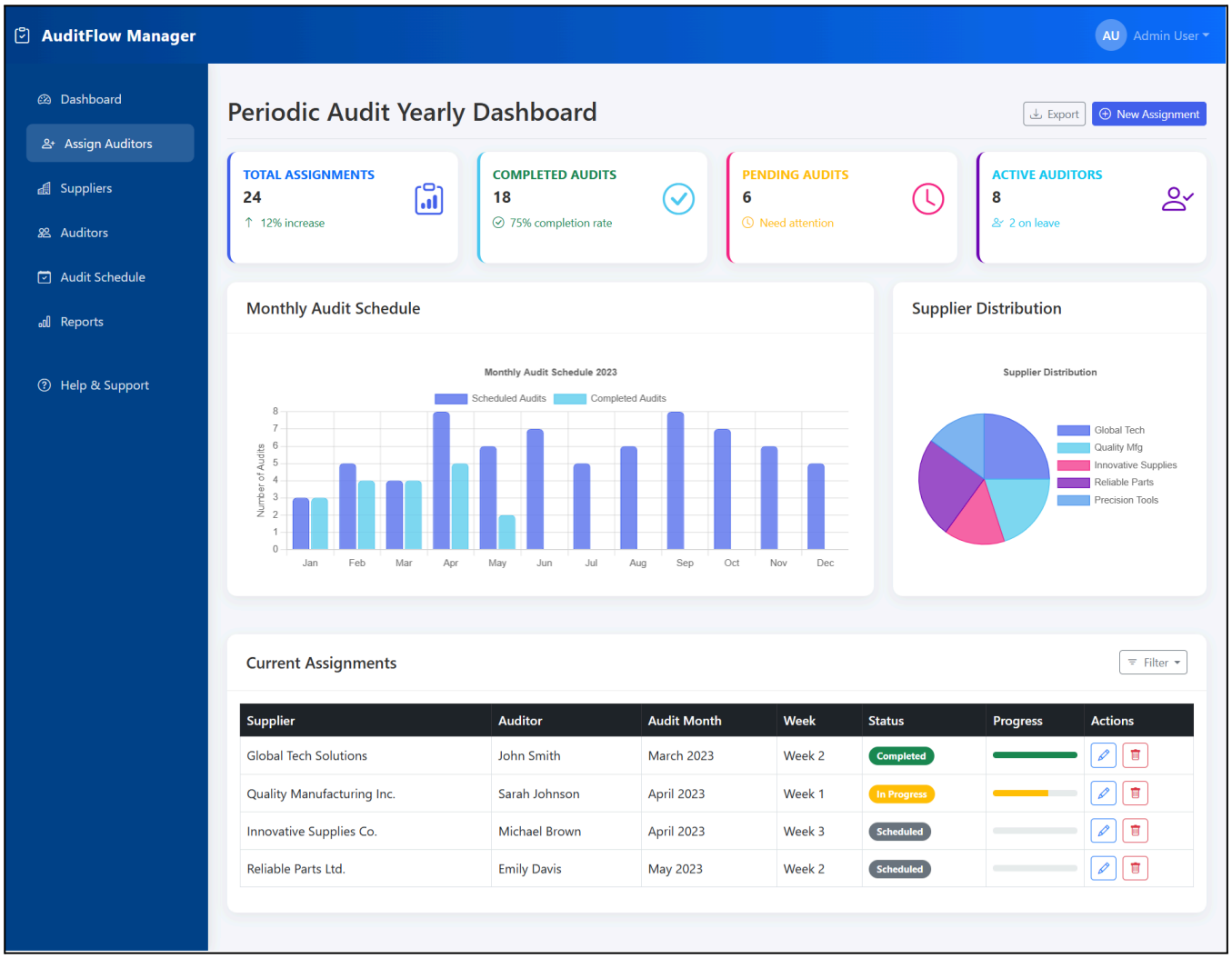
Inspection Remarks

Add any additional observations or notes about the dimensional inspection...

Submit Inspection

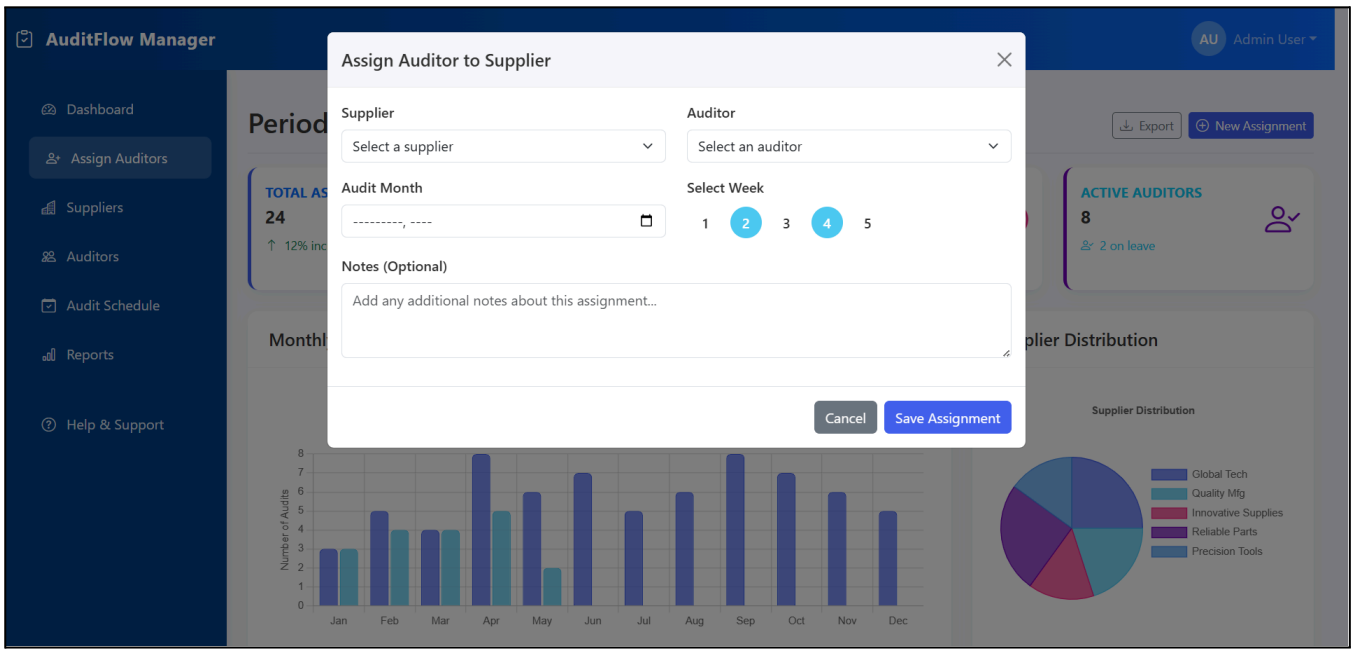
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3. Audit Dashboard

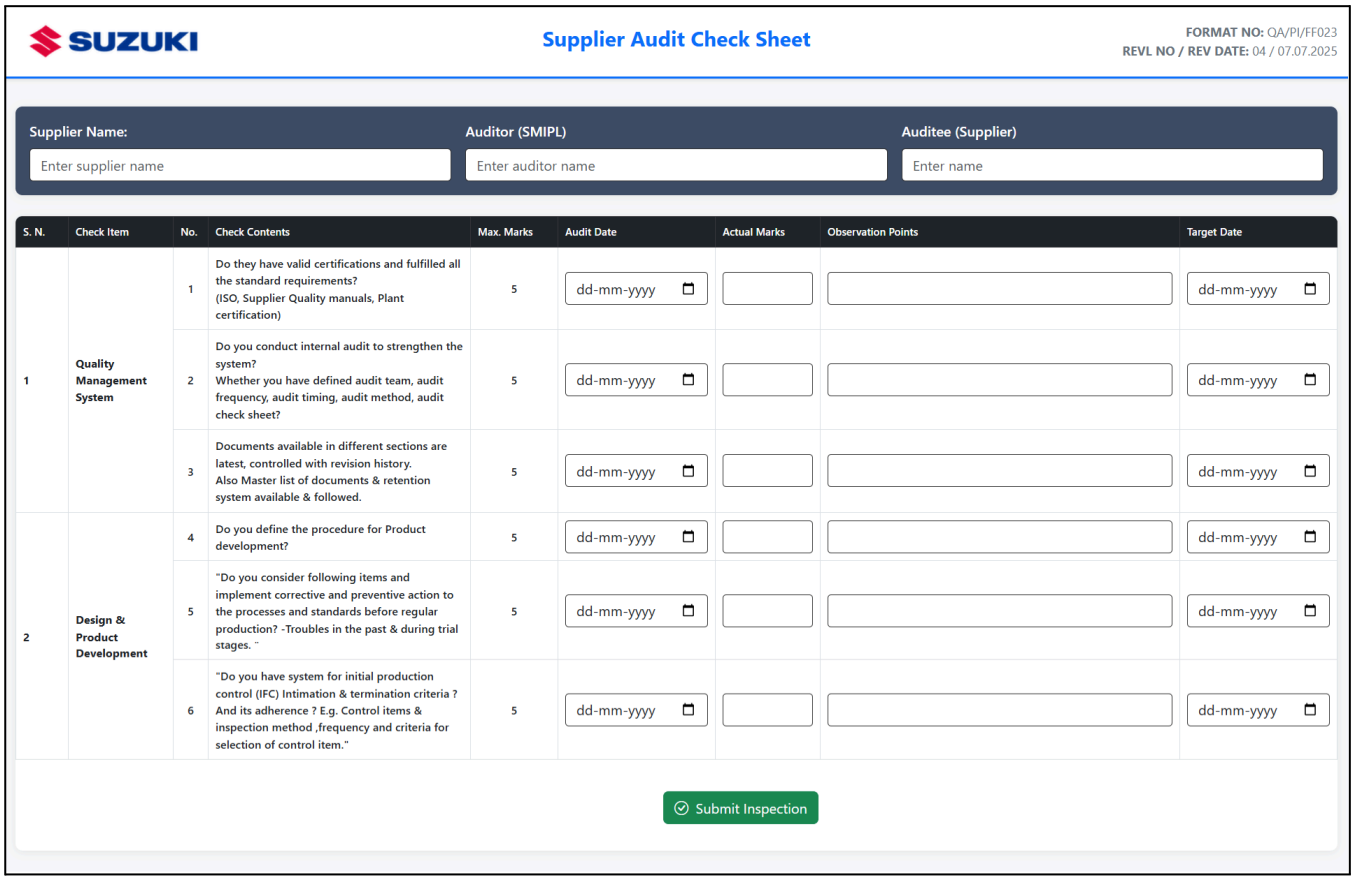


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4. Scheduling New Audit Plan



5. Supplier Audit Checksheet



6. SIS-P

SUZUKI SIS - P

PART CLASSIFICATION A PART AB PART	**MODEL NAME** XF325	**PAGE** 1 OF 6
FUNCTIONAL PART GENERAL PART	**SUPPLIER NAME** INDO AUTOTECH LIMITED	**SUPPLIER CODE** 3063
SUPPLIER INSPECTION STANDARD	**DATE OF ISSUE (FIRST ISSUE)** 15 NOVEMBER 2021	**STANDARD CLASSIFICATION** P

S I S	**REVISION NO** 0	**PART NAME** PIPE COMP FRAME HEAD	**PART NUMBER** 41110-02130-000	**RECEIVING CHECK** A

S NO	INSPECTION	JUDGEMENT CRITERIA	INSPECTION METHOD	INSPECTION TOOL	SAMPLING PLAN	INSPECTION STAGE
1	APPEARANCE	INNATURAL CRACK, BURL, VILLONIE DEFECTS UNDER CUT WIN HOLES CHARACTER ON BORING AREA 0.5 ETC. SHOULD NOT EXIST	VISUAL	-	100%	FINAL INSPECTION
		VILLONIE LOCATION / LEVERT AS FIRE DRAWING Annex-NTE-I	VISUAL	-	100%	FINAL INSPECTION
		FITMENT SHOULD BE OK	INSPECTION GAUGE		100%	FINAL INSPECTION
2	DIMENSIONS (in mm)					
1	DIMENSION Ø52.00 +0.074/0		3-PIN MICROMETER / AR-FLUG GAUGE		n=5/QT,c=0	FINAL INSPECTION
2	DIMENSION Ø45.00 +0.062/0		3-PIN MICROMETER / AR-FLUG GAUGE		n=5/QT,c=0	FINAL INSPECTION
3	DIMENSION 76.40 ±1.00		CAMA		n=5/week,c=0	FINAL INSPECTION
4	DIMENSION 321.00 ±0.80		HEIGHT GAUGE		n=5/QT,c=0	FINAL INSPECTION
5	DIMENSION 36.10 ±1.00		HEIGHT GAUGE		n=5/QT,c=0	FINAL INSPECTION
6	DIMENSION 225.90 ±1.00		HEIGHT GAUGE		n=5/QT,c=0	FINAL INSPECTION
7	DIMENSION 278.10 ±0.8		HEIGHT GAUGE		n=5/QT,c=0	FINAL INSPECTION
8	DIMENSION 38.00 ±0.50		HEIGHT GAUGE		n=5/QT,c=0	IN-PROCESS
9	DIMENSION 27.00 ±0.50		HEIGHT GAUGE		n=5/QT,c=0	IN-PROCESS
10	DIMENSION 12.00 ±0.80		HEIGHT GAUGE		n=5/QT,c=0	FINAL INSPECTION
11	ANGLE [xs]		CAMA		n=5/week,c=0	FINAL INSPECTION
12	DIMENSION 10.00 ±1.00		HEIGHT GAUGE		n=5/QT,c=0	FINAL INSPECTION
13	DIMENSION(CD) 55.00 ±0.50		VERNIER CAUFER		n=5/QT,c=0	FINAL INSPECTION
14	DIMENSION 112.00 ±1.60		VERNIER CAUFER		n=5/QT,c=0	FINAL INSPECTION
15	DIMENSION 39.60 ±2.00		CAMA		n=5/week,c=0	FINAL INSPECTION
16	DIMENSION 51.20 ±2.00		CAMA		n=5/week,c=0	FINAL INSPECTION
17	DIMENSION 75.00 ±2.00		CAMA		n=5/week,c=0	FINAL INSPECTION
18	DIMENSION 42.50 ±2.00		CAMA		n=5/week,c=0	FINAL INSPECTION
19	DIMENSION 86.30 ±2.00		CAMA		n=5/week,c=0	FINAL INSPECTION
20	DIMENSION 106.40 ±2.00		CAMA		n=5/week,c=0	FINAL INSPECTION
21	DIMENSION 139.00 ±1.00		CAMA		n=5/week,c=0	FINAL INSPECTION
22	DIMENSION 268.80 ±0.80		CAMA		n=5/week,c=0	FINAL INSPECTION
23	DIMENSION 94.40 ±1.00		CAMA		n=5/week,c=0	FINAL INSPECTION
24	DIMENSION 396.60 ±0.80		CAMA		n=5/week,c=0	FINAL INSPECTION
	DATA FORM	INSPECTION CHECCSHEET				
	REMARKS	Deviated dimension				

SUPPLIER _____ **CENTRAL OFFICE** _____

APPROVED BY: _____ **APPROVED BY:** _____

CHECKED BY: _____

PREPARED BY: _____

MARKS REASON FOR REVISION **PERSON IN CHARGE DATE**

MARKS	REASON FOR REVISION	PERSON IN CHARGE	DATE
	Dimension 82.5 ±1.5mm added and correction	Rajeev Ranjan	22.11.2022

PAGE 1 OF 6

7. SMIR

Summary of Monthly Inspection Reports

Vendor Name: NEO AUTOTECH LTD.				**Vendor Code:** 3063			**SMIPL Part Name:** PIPE COMP FRAME HEAD			**Date:** 30.11.2025		
SMIPL Part No: 41110-02L30				**Part Category:** Maru A			**Model:** XF375			**Format No:** QCE/PI/F/020		
S.No	**Inspection Item**	**Specification**	**Inspection Method**	**Sampling Plan**	**Apr-25**	**May-25**	**Jun-25**	**Jul-25**	**Aug-25**	**Sep-25**	**Oct-25**	**Nov-25**
I	APPEARANCE											
1	Harmful crack, burr, welding defect	Should not exist	Visual	100%	OK	OK	OK	OK	OK	OK	OK	OK
2	Welding location / length	As per DRG	Visual	100%	OK	OK	OK	OK	OK	OK	OK	OK
3	Acceptance as per funct. gauge	All pins & filler should be quality	Functional Gauge	100%	OK	OK	OK	OK	OK	OK	OK	OK
II	DIMENSIONS (IN MM)											
1	DIA	52.0 ±0.074	3-Pin Micrometer	n=5	52.047	52.047	52.045	52.045	52.045	52.047	52.047	52.045
2	DIA	45.0 ±0.002	3-Pin Micrometer	n=5	45.041	45.041	45.038	45.038	45.039	45.038	45.039	45.038
3	DIA	52.0 ±0.074	3-Pin Micrometer	n=5	52.047	52.047	52.045	52.045	52.045	52.047	52.047	52.045
4	DIA	45.0 ±0.002	3-Pin Micrometer	n=5	45.041	45.041	45.038	45.038	45.039	45.038	45.039	45.038
5	DIA	52.0 ±0.074	3-Pin Micrometer	n=5	52.047	52.047	52.045	52.045	52.045	52.047	52.047	52.045
6	DIA	45.0 ±0.002	3-Pin Micrometer	n=5	45.041	45.041	45.038	45.038	45.039	45.038	45.039	45.038
7	DIA	52.0 ±0.074	3-Pin Micrometer	n=5	52.047	52.047	52.045	52.045	52.045	52.047	52.047	52.045
8	DIA	45.0 ±0.002	3-Pin Micrometer	n=5	45.041	45.041	45.038	45.038	45.039	45.038	45.039	45.038
9	DIA	52.0 ±0.074	3-Pin Micrometer	n=5	52.047	52.047	52.045	52.045	52.045	52.047	52.047	52.045
10	DIA	45.0 ±0.002	3-Pin Micrometer	n=5	45.041	45.041	45.038	45.038	45.039	45.038	45.039	45.038
11	DIA	52.0 ±0.074	3-Pin Micrometer	n=5	52.047	52.047	52.045	52.045	52.045	52.047	52.047	52.045
12	DIA	45.0 ±0.002	3-Pin Micrometer	n=5	45.041	45.041	45.038	45.038	45.039	45.038	45.039	45.038
13	DIA	52.0 ±0.074	3-Pin Micrometer	n=5	52.047	52.047	52.045	52.045	52.045	52.047	52.047	52.045
14	DIA	45.0 ±0.002	3-Pin Micrometer	n=5	45.041	45.041	45.038	45.038	45.039	45.038	45.039	45.038
15	DIA	52.0 ±0.074	3-Pin Micrometer	n=5	52.047	52.047	52.045	52.045	52.045	52.047	52.047	52.045
16	DIA	45.0 ±0.002	3-Pin Micrometer	n=5	45.041	45.041	45.038	45.038	45.039	45.038	45.039	45.038
17	DIA	52.0 ±0.074	3-Pin Micrometer	n=5	52.047	52.047	52.045	52.045	52.045	52.047	52.047	52.045
18	DIA	45.0 ±0.002	3-Pin Micrometer	n=5	45.041	45.041	45.038	45.038	45.039	45.038	45.039	45.038
19	DIA	52.0 ±0.074	3-Pin Micrometer	n=5	52.047	52.047	52.045	52.045	52.045	52.047	52.047	52.045
20	DIA	45.0 ±0.002	3-Pin Micrometer	n=5	45.041	45.041	45.038	45.038	45.039	45.038	45.039	45.038
III	PERFORMANCE / ENDURANCE											
1	Penetration Test	15% Min	Micro Scope	n=1	OK (AS PER WPT REPORT)							
2	Torque Test	16 Nm Min	Torque Wrench	n=1	OK (AS PER WPT REPORT)							
IV	MATERIAL											
1	PIPE FRAME HEAD	STKM11A / ERW-1	Lab Report	n=1	OK (AS PER MTC REPORT)							
2	PLATE STEERING STOP	JSH270CN-PHROP	Lab Report	n=1	OK (AS PER WPT REPORT)							

8. PACD

PACD (Problem Analysis and Correction Document)

RECEIVED – SUZUKI

FORMAT NO : QCE/PI/F/006		**REVISION NO & DATE : 01 / 01.04.2018**	
Issuance No	B2509001	**Date of Issuance**	03.09.2025
Supplier Name	M/s SURI AUTO		
Part Name	NUT	**Part No**	53122-27L00-000
Model	XF3P*	**Place of Occurrence**	SMIPL Line - 3
Date of Occurrence	03.09.2025	**Date of Receiving**	17.09.2025

1. Defective Content and Occurrence Situation

Nut front fender thread operation found miss.
Without thread operation part found during packing process.

Classification of Parts	**☑ Domestic**	**☑ Export**	**☐ Others**

2. Investigation Result

No. of Occurrence	140 Nos	Lot No	N/A
Disposition / Status	140 Nos parts rejected and returned to supplier		

3. Causes of the Defects

Helper mixed rework bin parts and OK bin parts and sent for packing.

4. Causes of Outflow

Defect not detected during PDI due to less sample size for thread checking.

5. Preventive Countermeasures

1. 100% parts checking through TPG.
2. Training to inspectors & operators.
3. Bin color coding implemented.
4. Sensor-based checking machine planned.

7. Effectiveness of Countermeasures

Total Checked	8000	NG	140
OK	7860 Nos		

Treatment Dept	**Quality**	**Responsible**	**Prabhakar Dabral**
Approval	Pravanjan Ray		
Delivery Time	**Next Lot**	**Delivery Place**	**SMIPL**

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E. Technology Platform Recommended by Technology Services Provider

The development of the solution will be compatible with all the latest browsers except Safari.

Platform	Technology
Backend Development	ASP .NET Core
Frontend Development	Angular
Android Development	Native Android
Database	Microsoft SQL Server
UI	HTML, CSS, JavaScript, Bootstrap
Hosting	Azure
Security	HTTPS, JWT-based authentication Role-based access control

Technology Services Provider will require remote access for all server environments. Production, Development and QA servers hosted on the cloud will be provisioned, maintained and cost-covered by Global Automotive Manufacturer. If required by Global Automotive Manufacturer to support the cloud operations by Technology Services Provider it will be charged separately.

The following configuration will be required for the development and deployment of the application:

1. Application Server (Angular + ASP.NET Core)

- CPU: 4-core
- RAM: 16 GB DDR4 ECC
- Storage: 100 GB SSD (OS, Angular builds, logs)
- OS: Windows Server 2022
- Deployment: IIS
- Backup : Daily Incremental + Weekly Full

2. Database Server (SQL Server)

- CPU: 4-core high-frequency
- RAM: 24 GB ECC
- Storage: 250 GB SSD (50 GB OS, 20 GB logs, 150 GB DB, 30 GB backups)
- SQL Edition: SQL Server 2019/2022 Standard or Enterprise
- OS: Windows Server 2022 Standard
- Backup : Daily Incremental + Weekly Full

3. QA Server (Angular + ASP.NET Core + SQL Express Edition)

- CPU: 4-core
- RAM: 24 GB DDR4 ECC
- Storage: 200 GB SSD (OS, Angular builds, logs, SQL Express)
- OS: Windows Server 2022
- Deployment: IIS
- DB Engine : SQL Server 2022 Express Edition
- Backup : Daily Incremental + Weekly Full

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4. **Development Server (Angular + ASP.NET Core + SQL Standard Edition)**

- User Access: 3 users
- CPU: 6-core
- RAM: 32 GB DDR4 ECC
- Storage: 500 GB SSD (OS, Angular builds, logs, SQL Express)
- OS: Windows Server 2022
- Deployment: IIS
- DB Engine : SQL Server 2022 Standard Edition
- Backup : Daily Incremental + Weekly Full

5. **Software & Licensing**

- Windows Server: 2x Standard licenses
- SQL Server: 2019/2022 Standard or Enterprise
- Antivirus: Any

6. **Additional Details**

- Server Availability : 24*7
- Public IP
- SSL Certificate
- Antivirus
- SMTP

F. Tentative Timelines

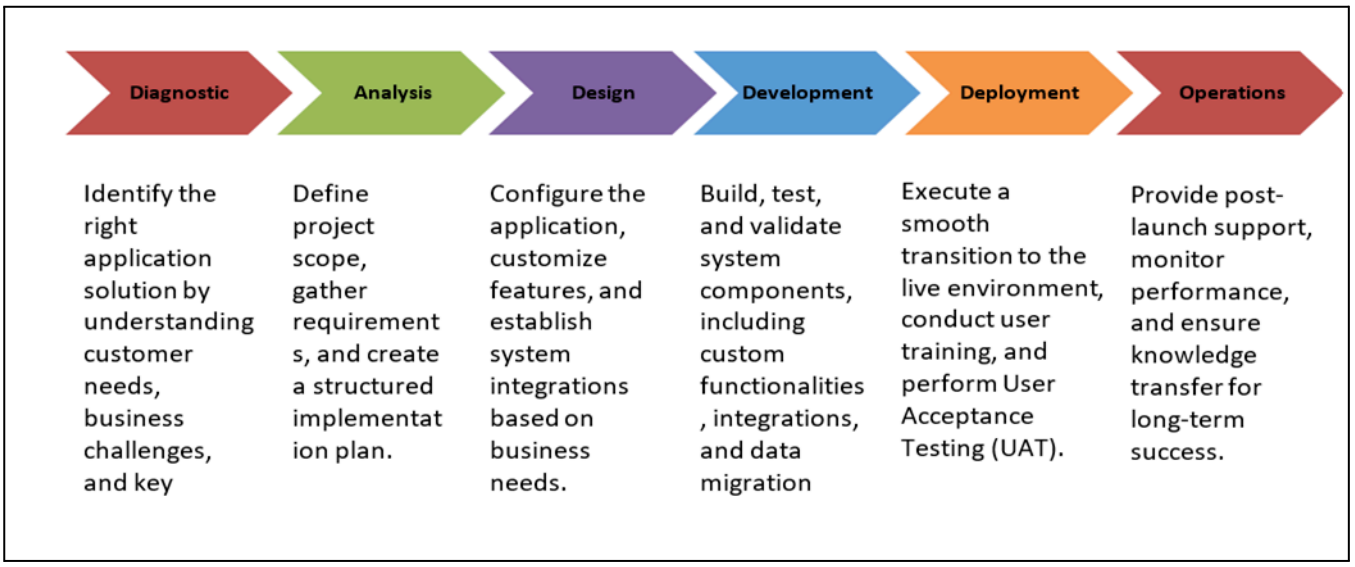
This is subject to refinement based on BRD/FRD Sign-Off.

Data Management System Development Timelines																									
Milestone	Weeks																								
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
BRD/Project Documentation Preparation & Sign Off																									
UI Designs																									
Development & Integration																									
QA Testing (Use Cases)																									
UAT Deployment																									
UAT Test & Feedback Incorporation																									
Project Documentation (User Manual, Technical Documentation)																									
UAT Feedback Reverification & Go Live Preparation																									
Project Go Live																									

- Hypercare will be given for 3 months immediately after project go-live with the expectation of as and when required.

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G. Implementation Methodology



Please Note: The above is the generic implementation methodology with reference to our current document, the phases of this methodology would be executed as applicable.

H. Change Request (CR) Management

1. Change Request (CR)

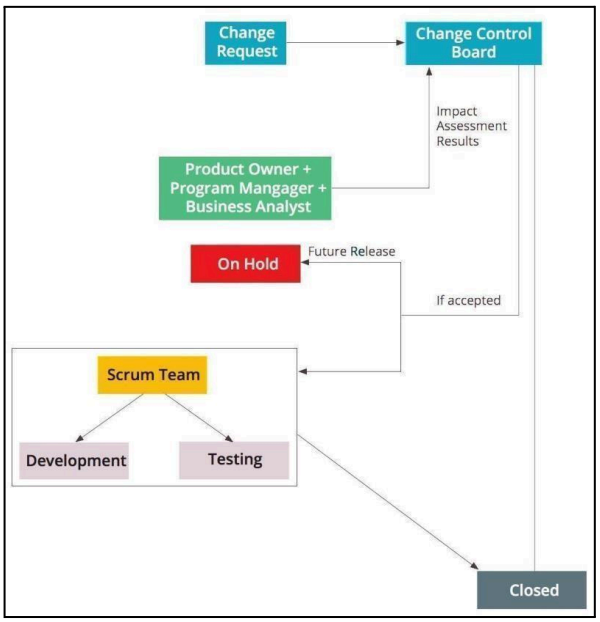
Technology Services Provider proposes that changes to requirements be handled using the project change request management tools and procedures duly signed off by the client and the Technology Services Provider Project in-charge. The major stakeholders of the project need to have a clear understanding of these changes and the associated implications on the project effort, timelines, cost, and related systems. Technology Services Provider along with client project managers and business stakeholders will work effectively to absorb variations from initial functional requirements and discuss trade-offs to balance between scope, schedule, and resources (budget), if possible.

If the Global Automotive Manufacturer Team wishes to introduce different goals or principles based upon new/added factors introduced or discovered by the teams, including, but not limited to, the type of work to be done, the number of hours or work to be performed, the schedule estimated, or a change in management. Direction affecting the current scope of the Project, or any other changes in direction, priority, schedule, or approach once established and agreed to in this proposal, must undergo this CR process. Such changes will necessarily have both a schedule and cost impact on the Project and these changes will not be binding upon Technology Services Provider within the current scope shall have both an equitable price and schedule adjustment. Any such Change Requests duly signed off, will be executed by Technology Services Provider on an additional time, cost, and effort basis.

2. CR Management Approach

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The goal of the change management process is to produce a more robust system that better supports the business activities of the clients. The change management process
includes –



- **Collect** change requests (a generic term that encompasses defects, bugs, requests, and suggestions). This is a continuous process and will be done during each sprint sign-off.
- Identify the scope of the next release and determine which change requests will be included in the next build. The change requests should be grouped according to the functional and/or physical area of the system the request will change.
- **Document** the requirements, functional specifications, and implementation plans for each grouping of change requests within the Product/Project with a proper sign-off
- **Implement the change** This may involve changes to source code, database structure, or user documentation.
- **The test** is to verify the changes done by the person who made the change, usually the programmer. It is very specific to just the area changed. Tests a functional area of the system to see that everything works as expected.
- **Release** the system to Product/Projection. This is the physical process of upgrading the Product/Projection system to the new release. This may involve executable files, database changes, data migration, and the installation of new support software.
- **Change** Management Impact could be a variance to the budgeted efforts, timelines, and cost of the original project scope of work.

3. Definition of Change Request

The following will be categorized and treated as Change Requests

1	Any specific change in scope item beyond what is defined in the BRD, requiring an effort of more than 2 man-days
2	Document sign-off, i.e., BRD taking more than 2 working days from submission.
3	Post UAT. The UAT sign-off takes more than 2 working days.
4	Any delays caused by the unavailability of data that lead to a pause in the project for more than 2 days.
5	Any delays caused due to issues in availability/ access to Infra and/or the customer's internal core systems for more than 2 days.

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I. Dependencies, Assumptions, Out of Current Scope

A. Dependencies on the Global Automotive Manufacturer Team

1. Finalization of SIS-P and SMIR workflows, including creation, submission, validation, and approval processes.
2. Provision of the SIS-P approval matrix by Global Automotive Manufacturer, defining all stakeholders, roles, and escalation levels.
3. Confirmation of PACD workflows, including supplier approvals, Global Automotive Manufacturer approval hierarchy, and closure criteria.
4. Agreement on SLAs such as PACD submission timelines (15 days), SMIR cycles, and audit timelines.
5. Availability of standardized checksheet formats, validation rules, and business logic to be implemented in the system.
6. Access to all legacy SIS-P PDF documents (~4000 records) required for OCR-based data migration.
7. Sharing of sample SIS-P and SMIR documents for training and validating OCR extraction.
8. Final definition of data fields to be extracted from SIS-P and SMIR and stored in the database.
9. Final list of user roles (PI Team, Supplier SPOC, Supplier Approver, Global Automotive Manufacturer Approvers, Admin) and their responsibilities.
10. Availability of supplier master data including supplier codes, SPOCs, and approver mappings.
11. Technology Services Provider will leverage the existing Private and Public domain URLs along with the TLS/SSL certificates currently used in Phase 1 of the PI DMS for hosting Phase 2. Branding guidelines (logos, color schemes, fonts, and other branding elements).
12. Global Automotive Manufacturer will provide a subscription for any required third-party service.
13. Infrastructure for Testing/UAT and Production is to be provided by the Global Automotive Manufacturer Team infrastructure team.
14. High-resolution images, product photos, team member portraits, and other relevant imagery are essential for a professional appearance.
15. Third-party API Integrations like API keys, access credentials, and integration documentation
16. All the network infrastructure, firewall, and configuration of private/public IPs and SSL certificates will be provided by the Global Automotive Manufacturer infrastructure team.
17. All meetings should happen on time without any delay.
18. The Global Automotive Manufacturer Team needs to ensure the availability of testing devices, the availability of users, and sample data at the time of testing and training including test cases and scenarios.
19. Test cases and scenarios should be provided and frozen at the time of the requirement-gathering phase.
20. Test cases and scenarios once decided will not change.
21. Any deliverables (e.g. approvals, UATs, sign-offs, etc.) submitted to the Global Automotive Manufacturer Team should be reviewed and signed off within 10 working days of delivery (since this is a tightly time-bound project), else it will be deemed accepted or approved by the end of 10th day. If there is any delay due to Global Automotive Manufacturer Team

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dependencies, the additional days required would be serviced based on effort and cost basis.

22. Any approval provided by email will be considered formal approval and sign-off.
23. The UI design will be created by Technology Services Provider, and feedback on the design will be provided by the Global Automotive Manufacturer team.
24. For Notification mail server details to be provided by the Global Automotive Manufacturer Team.
25. All the integration credentials & access are to be provided by the Global Automotive Manufacturer Team.
26. The email content will be designed by Technology Services Provider and feedback will be provided by the Global Automotive Manufacturer team over it.
27. The email content will be designed by Technology Services Provider and feedback will be provided by the Global Automotive Manufacturer team.

B. Assumptions

1. The Android application will be limited to supporting the Parts Inspection functionality only.
2. The SIS-P, SMIR, and PACD processes defined during discussions will remain largely stable during the development phase, with no major scope changes post sign-off.
3. Global Automotive Manufacturer will provide the final approval matrix, workflows, and business rules in a timely manner to avoid design or development delays.
4. All legacy SIS-P documents (~4000 PDFs) will be available, readable, and suitable for OCR processing without significant quality issues.
5. To host the mobile application on Google Play Store, Global Automotive Manufacturer will provide the Google Play Developer Account along with the required details to procure and manage the account, including the associated Google account.
6. Global Automotive Manufacturer will provide timely access to key stakeholders, subject matter experts, and decision-makers for requirement validation and UAT sign-offs.
7. Any enhancement or change after BRD sign-off will be treated as a Change Request (CR) and estimated separately.
8. The project duration and delivery are based on mutual dependencies and timely approvals from Global Automotive Manufacturer.
9. Global Automotive Manufacturer will nominate business users and approvers to perform UAT within defined timelines.
10. UAT feedback will be consolidated and shared formally for closure - repeated UAT cycles due to scope additions will be treated as change requests.
11. Infrastructure for development, testing, and production is to be provided by the Global Automotive Manufacturer Team.
12. Wireframes will be prepared by Technology Services Provider and will be shared with Global Automotive Manufacturer SPOC for approval.
13. Global Automotive Manufacturer will ensure effective communication of Technology Services Provider and Partner Ecosystem for necessary integration.
14. UAT and Development will be done in the customer environment.
15. Any repeat work undertaken by Technology Services Provider, due to miss/failure by Global Automotive Manufacturer Team time will be executed on an additional cost, time & effort basis.
16. The customer needs to nominate specific SPOCs for all UATs. All UATs are restricted to be executed only by the above SPOCs and not by any other observer of the project.

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17. The Global Automotive Manufacturer Team IT will be SPOC for this project and will communicate with the business team.
18. Hyper care will be initiated after the post-live and expiry of the hyper care the support contract needs to be signed for the support on an additional cost, time & effort basis.
19. The post-Go-live Portal admin activity is to be performed by the Global Automotive Manufacturer Team.
20. The UAT will be limited to the test cases defined at the requirement gathering phase.
21. All the notification email template content will be provided by the Global Automotive Manufacturer Team.
22. Mail Server (SMTP) details or any third-party mail services to be provided by the Global Automotive Manufacturer.
23. The Global Automotive Manufacturer Team is to share the designated SPOC for this project preferably from their IT and Business Teams.
24. The portal will be accessed through the latest version of the web browser only except the Safari browser.
25. Hyper care will be initiated after the post-live and on expiry of the hyper care, the annual support contract needs to be signed for any support on the delivered solution, on an additional cost, time & effort basis.

C. Out of Current Scope

1. Development of the iOS application is not part of the scope.
2. Creation or editing of 4M diagrams within the system (only image upload will be supported).
3. Data cleansing, transformation, or validation of past data.
4. Changes to VRMS/CDSS systems or development of new external APIs beyond the interfaces provided by Global Automotive Manufacturer.
5. Data correction or cleansing of legacy SIS-P records beyond basic OCR extraction and structured storage.
6. Provisioning and maintenance of servers, databases, network access, VPNs, or cloud infrastructure.
7. Any third-party integration for notifications (e.g., Teams, WhatsApp, or Slack).
8. Training or process rollout to end-users across departments beyond the initial train-the-trainer sessions.
9. Technology Services Provider will not be responsible for data correctness, completeness, or duplication across SAP or other systems.
10. Validation and reconciliation of financial data remain Global Automotive Manufacturer's responsibility.
11. Multilingual interface support (English-only UI is assumed).
12. Offline / desktop-based version of the DMS system.
13. Third-party license costs (if required, will be billed at actuals).
14. The system will provide a mobile-responsive web interface for the application, accessible via web browsers only.
15. Installation, configuration, or licensing of SQL Server, Windows Server, IIS, SSL certificates, or other system software.
16. Backup and disaster recovery setup beyond standard deployment support.
17. Performance monitoring tools or dashboards at server level (handled by Global Automotive Manufacturer IT).

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18. Any new process, new functionality, new application, new workflows, etc. not mentioned in the scope. The same would be catered through the Change Request (CR) process on a Time, Material, and cost basis if this new scope is relevant to this project scope. If the new/added scope is directly relevant to the current scope and requires a fresh development, the same might be executed as a new project phase, based on a mutual agreement on the same.
19. Cost of any other software, infra., tools, etc. required for implementation and deployment of the application.
20. Any issues arising due to infrastructure and/ or reasons not in control of Technology Services Provider.
21. No third-party tool will be implemented/integrated into the portal unless otherwise already mentioned in this Scope document.
22. Management of hardware of the Database and Applications.

J. Project Communication and Control

A. Documentation

1. Once the project is awarded, Technology Services Provider will share the following –
    - Business Requirement Document (BRD)
    - Project Tracker
    - Technical Design document
    - User manual
    - Weekly Project Status Report
2. BRD which will need to be mutually signed off, is the basis on which the project will be executed.
3. A Project Tracker will be shared which will have detailed Sow-Tasks, Dependencies, Prerequisites, Ownership (both Global Automotive Manufacturer Team & Technology Services Provider), Task Status, ETC, and ATC and Remarks. This needs to be updated and shared on an agreed periodic basis.
    - This document needs to be monitored by both Technology Services Provider & Global Automotive Manufacturer Team to ensure a smooth and timely execution of all executables.
    - The Global Automotive Manufacturer Team is equally responsible for owning the progression tracking (especially for tasks owned by the Global Automotive Manufacturer Team.
    - Any delay by the Global Automotive Manufacturer Team in executing their tasks might impact the execution timelines and hence the cost.
4. Project Delivery Completion Document which will be required to be signed off by the Global Automotive Manufacturer Team along with the Go-Live sign-off.

B. Governance

1. The project review call/meeting is planned every week. The mode of the review can be direct or web whichever is agreed between the project managers of both divisions. The Project Status report (PSR) is sent to the stakeholders every week after the review meeting.
2. Governance will be done leveraging a Project Tracker and a periodic project execution review which will be attended by the Project SPOCs and relevant team members of both the Global Automotive Manufacturer and Technology Services Provider Team.

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3. A collaborative governance model for smooth IT operations, issue resolution, and escalation process for effective monitoring and quality control of project execution is followed as a standard practice by Technology Services Provider.
4. This governance will lay an effective relationship management with an evolution of the account governance framework, as our relationship grows from a service provider to a trusted advisor. Such governance can also help in pro-active forecasting of clients' upcoming solution & professional resource needs, by gauging business requirements, operations & supporting domain functions.
5. The general practice followed by Technology Services Provider to facilitate all of the above is as depicted below.

K.

Governance Category	Frequency	Governance Members
PSR (Project Status Report)	Once every Week	Project Task Owners / Project SPOCs
Project Track Meetings	Fortnightly	Project Managers & Delivery Head

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